untitled

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```
from collections import deque
from math import gcd

def water_jug(cap_a, cap_b, target):
    if target > max(cap_a, cap_b) or target % gcd(cap_a, cap_b) != 0:
        return None
    start = (0, 0)
    parent = {start: None}
    queue = deque([start])
    while queue:
        a, b = queue.popleft()
        if a == target or b == target:
            path = []
            state = (a, b)
            while state is not None:
                path.append(state)
                state = parent[state][0] if parent[state] else None
            path.reverse()
            return path, parent
        pour_ab = min(a, cap_b - b)
        pour_ba = min(b, cap_a - a)
        options = {
            "Fill A": (cap_a, b),
            "Fill B": (a, cap_b),
            "Empty A": (0, b),
            "Empty B": (a, 0),
            "Pour A -> B": (a - pour_ab, b + pour_ab),
            "Pour B -> A": (a + pour_ba, b - pour_ba),
        }
        for action, nxt in options.items():
            if nxt not in parent:
                parent[nxt] = ((a, b), action)
                queue.append(nxt)
    return None

def print_solution(cap_a, cap_b, target):
    print(f"Jugs: {cap_a} L and {cap_b} L, target = {target} L")
    result = water_jug(cap_a, cap_b, target)
    if result is None:
        print("No solution possible.\n")
        return
    path, parent = result
    print(f"{'Step':<5}{'Action':<12}{'(A, B)'}")
    print(f"{0:<5}{'Start':<12}{path[0]}")
    for i, state in enumerate(path[1:], 1):
        print(f"{i:<5}{parent[state][1]:<12}{state}")
    print(f"Solved in {len(path) - 1} steps.\n")
```

Ln: 52 Col: 47

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IDLE Shell 3.11.1

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```
7 _ 8

Step 8: move blank Right
1 2 3
4 5 6
7 8 _

Solved in 8 moves.

>>>
========== RESTART: C:/Users/ASUS/pyhon.py ==========
First solution (row -> column): [1, 5, 8, 6, 3, 7, 2, 4]
Q . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .

Total number of solutions for 8 queens: 92

>>>
========== RESTART: C:/Users/ASUS/PY.py ==========
Jugs: 4 L and 3 L, target = 2 L
Step Action      (A, B)
0    Start       (0, 0)
1    Fill B      (0, 3)
2    Pour B -> A (3, 0)
3    Fill B      (3, 3)
4    Pour B -> A (4, 2)
Solved in 4 steps.

Jugs: 5 L and 3 L, target = 4 L
Step Action      (A, B)
0    Start       (0, 0)
1    Fill A      (5, 0)
2    Pour A -> B (2, 3)
3    Empty B     (2, 0)
4    Pour A -> B (0, 2)
5    Fill A      (5, 2)
6    Pour A -> B (4, 3)
Solved in 6 steps.

Jugs: 6 L and 4 L, target = 5 L
No solution possible.

>>>
```

Ln: 89 Col: 0

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